



MRC
Biostatistics
Unit



UNIVERSITY OF
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Bayesian methods for clinical trials

Lecture 1: Introduction to Bayesian Statistics

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Situations where frequentist methods struggle

The p -value answers *how surprising are these data if the drug does nothing?*

Real question often *how likely is benefit, and how large?*

- “Not statistically significant” gets read as “no benefit” — confusing **absence of evidence with evidence of absence**.
- External information (adult data, prior device generations, earlier trials) is deemed inadmissible

Example: EOLIA: A “negative” trial that was not

EOLIA (Combes et al., 2018): ECMO for very severe ARDS, $n = 249$, stopped early for futility.

	60-day mortality	Frequentist verdict
ECMO	35%	RR 0.76, 95% CI 0.55–1.04
Control	46%	$p = 0.09 \Rightarrow$ “no benefit”

Bayesian re-analysis of the same data (Goligher et al., 2018):

- Posterior probability of *any* mortality benefit: roughly 88%–99%, across priors from strongly skeptical to enthusiastic.
- Same data, same investigators — what changed is reporting effect size and uncertainty instead of a pass/fail verdict.

Same pattern in ANDROMEDA-SHOCK ($p = 0.06$): posterior probability of benefit above 90% (Zampieri et al., 2020).

Challenge: Small populations

Rare disease, paediatrics and devices share one problem: the required sample size not achievable.

- **Pediatric populations.** Often few children or ethically challenging. A traditionally powered trial often infeasible.
- **Rare diseases.** Rarity of condition means traditional sample sizes would require years or decades to recruit.
- **Devices.** Infeasible to run next generation device studies to same standard and size as initial device study.

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Solution

Information borrowing

Note: The frequentist alternative is rarely “no borrowing” — it is informal borrowing.

Examples: Small populations

Three products licensed on Bayesian evidence:

- **REBYOTA** (Khanna et al. 2020, Clostridioides difficile Infection), FDA approval in 2022. Pre-specified primary analysis was a Bayesian hierarchical model dynamically borrowing from the phase 2b PUNCH CD2 trial. Posterior probability of superiority 99.1% vs. a 97.5% threshold.
- **Optimizer Smart** (Abraham et al, 2018, cardiac contractility modulation), FDA approval in 2019. FIX-HF-5C, $n = 160$, pre-specified Bayesian borrowing of peak VO_2 data from the earlier FIX-HF-5 trial; mean difference 0.84 mL/kg/min (95% CrI 0.12–1.55).
- **PLUTO** (Brunner et al, 2020; Pottackal et al. 2025, childhood-onset SLE), FDA approval in 2019. PLUTO randomised 93 children and was underpowered by design. FDA's own analysis used a robust mixture prior based on the adult BLISS trials — and any weight $\geq 55\%$ on the adult data gave a 95% credible interval excluding $OR = 1$.

- Response adaptive randomisation (more common now in platform trials) often Bayesian.
- Incoherent to use different framework for analysis
- Examples
 - ▶ **I-SPY 2** (neoadjuvant breast cancer)
 - ▶ **REMAP-CAP** (COVID-19)
 - ▶ **ARREST** (ECPR for refractory VF arrest)

What Bayes does *not* fix

- **Priors are assumptions, not free evidence.** Need to be pre-specified and well justified.
- **Prior–data conflict is real.** Borrowed adult or historical data bias the estimate if the effect truly differs.
- **Operating characteristics are still required.** Type I error and power often come from simulations

Important to remember

Bayesian methods do not yield a better chance of a positive trial. It results in a report that says how large the effect probably is, how sure we are, and how much of that rests on what we assumed beforehand.

Bayesian methods: Toy Example

- 15 people who reported a headache were given aspirin.
- 10 reported their headache vanishing within half an hour.

What inferences can we make about **probability of response**?

Conventionally, the probability of response could be given by **expectation** $10/15 \approx 0.67$.

- Frequentist estimate could be

$$se = \sqrt{\frac{\hat{p}(1-\hat{p})}{N}} = \sqrt{\frac{0.67(1-0.67)}{15}} = 0.12$$

- The 95% confidence interval is (0.428,0.905).
- Problems with what the resultant CI means and how to use it.

A Bayesian Approach

Bayesian methods offer different solutions.

- Consideration of the parameter of interest, denoted by θ .
- In this case, θ is the probability of headache vanishing within half an hour given aspirin, as in itself a **random variable** that have a distribution (reflecting our uncertainty in it).
- Information about the nature of this random variable comes from the sample.

How do we go about calculating a suitable distribution for θ when θ is continuous variable taking values between 0 and 1?

- Let θ be the parameter of interest with prior distribution $\pi(\theta)$.
- The observations are $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$, and have a marginal density function $f(\mathbf{x})$.
- The likelihood function $f(\mathbf{x}|\theta)$ is the distribution of $\mathbf{x}$ conditional on specific values of θ .

Then:

$$\pi(\theta|\mathbf{x}) = \frac{f(\mathbf{x}|\theta)\pi(\theta)}{f(\mathbf{x})}$$

Note: π is used to denote functions of θ , and f functions of the data.

- The function $f(\mathbf{x})$ is the marginal distribution of the data.
- For most problems, we are not that interested in $\mathbf{x}$, but in θ .

Because of this we can write the posterior distribution as:

$$\begin{aligned}\pi(\theta|\mathbf{x}) &= \frac{1}{f(\mathbf{x})} \pi(\theta)f(\mathbf{x}|\theta) \\ &\propto \pi(\theta)L(\theta)\end{aligned}$$

as $f(\mathbf{x})$ can be regarded as a normalising constant.

Critical to Bayesian inference are likelihood functions. These

- serve to link the **sampling** space to the **parameter** space.

The probability:

$$f(\mathbf{x}|\theta) = L(\theta) = \prod_i f(x_i|\theta)$$

is a general expression for a likelihood function given *iid* data.

Example: Bernoulli likelihood

- The observation of 10 responses from 15 patients is a series of Bernoulli trials with $\mathbf{x} = \{0, 1\}$. For any single trial the probability mass function is

$$f(x_i | \theta) = \theta^{x_i} (1 - \theta)^{1-x_i}$$

- The likelihood function is therefore:

$$f(\mathbf{x} | \theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{\sum_i x_i} (1 - \theta)^{n - \sum_i x_i}$$

- The likelihood for a series of iid Bernoulli trials can be modelled:

$$L(\theta) = \theta^{\sum_i x_i} (1 - \theta)^{n - \sum_i x_i}$$

- As for a series of $\sum x_i$ where $\mathbf{x} = \{1, 0\}$ is the number of “successes”, and n the number of trials, then $n - \sum x_i$ can be seen as the number of “failures”. $\sum x_i$ can be denoted s .
- This is the functional form of the *Beta* distribution.
- θ is restricted such that $0 \leq \theta \leq 1$.

Consider the Beta prior distribution $\mathcal{B}(a, b)$ of the probability of response

$$\pi(\theta) = \frac{1}{\text{Beta}(a, b)} \theta^{a-1} (1 - \theta)^{b-1}$$

where $\text{Beta}(a, b)$ is Beta function.

The parameters of the prior distribution are called **hyperparameters** to distinguish them from the parameters of the sampling space.

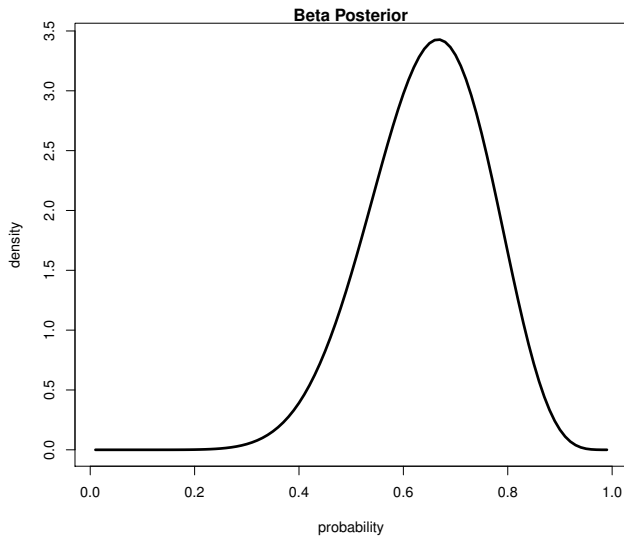
$$\begin{aligned}\pi(\theta|\mathbf{x}) &\propto \pi(\theta)f(\mathbf{x}|\theta) \\ &= c \left[\theta^{a-1}(1-\theta)^{b-1} \right] \times [\theta^s(1-\theta)^{n-s}] \\ &= c \left[\theta^{a-1}\theta^s(1-\theta)^{b-1}(1-\theta)^{n-s} \right] \\ &= c \theta^{a+s-1}(1-\theta)^{b+n-s-1}\end{aligned}$$

It follows that the posterior distribution is

$$\theta|\mathbf{x} \sim \text{Beta}(s+a, n-s+b)$$

The chosen Beta prior is called **conjugate prior**: A prior is conjugate to likelihood if posterior is of the same family as prior.

Beta Posterior



Summaries of the Beta distribution

- For the *Beta* distribution, there is a number of summaries which are readily available:

1.

$$E(\theta) = \frac{a + s}{a + b + n}$$

2.

$$\text{mode} = \frac{a + s - 1}{a + b + n - 1}$$

3.

$$\text{var}(\theta) = \frac{(a + s)(b + n - s)}{(n + a + b)^2(n + a + b + 1)}$$

Posterior means and variances are useful but don't tell us everything.

The posterior distribution can give as various sort of information about the distribution of the parameter.

A specific region which contains a given area of the posterior density function known as a **Credible Interval**.

- **HDR (highest density region):** smallest interval for the parameter of interest which contains $(1 - \alpha) \times 100\%$ of area.
- **Equal-Tailed Interval:** the interval where the probability of being below the interval is as likely as being above it.
- **Interpretation:** there is a probability of $(1 - \alpha) \times 100\%$ that the parameter of interest falls within the region.

Now, we would like to make a prediction. Assume that 10 more patients had headaches and were assigned aspirin.

What is the probability that 7 of them will respond?

$E(\theta) \approx 0.65$, so should we use the binomial distribution to compute it?

$$\Pr(X = 7) = \frac{10!}{k!(10 - 7)!} 0.65^7 (1 - 0.65)^{(10-7)}$$

```
dbinom(x=7,size=10,prob=11/17)  
[1] 0.2505501
```

This, does not take into account the **uncertainty** about θ .

In many cases, where the observations are assumed to be independent and identically distributed given θ , we have that

$$f_Y(y|\mathbf{x}) = \int f(y|\theta)\pi(\theta|\mathbf{x}) d\theta. \quad (1)$$

Posterior predictive distribution

The posterior predictive distribution is the distribution of possible unobserved values conditional on the observed values.

A posterior predictive distribution accounts for uncertainty about θ . So, instead of computing $\Pr(X = 7 | \theta = 0.65)$, we compute

$$\Pr(X = 7 | \text{data}) = \int_0^1 \Pr(X = 7 | \theta) \times \pi(\theta | \mathbf{x}) d\theta.$$

In our example, it is the **beta-binomial** distribution

$$\pi(k | N, \alpha, \beta) = \frac{\Gamma(N+1)}{\Gamma(k+1)\Gamma(N-k+1)} \frac{\Gamma(k+\alpha)\Gamma(N-k+\beta)}{\Gamma(N+\alpha+\beta)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}.$$

where $\alpha = s + a$ and $\beta = n - s + b$ (parameters of our Beta posterior) and N is the number of new patients.

$$\pi_{\text{pred}}(X = 7 | \text{data}) \approx 0.205.$$

```
library("rmutil")  
dbetabinom(7, 10, 11/17, 17)  
[1] 0.2050343
```

Using our previous estimate of the probability, we overestimate the probability, as we it does not take into account that a lot of probability mass lies below $\theta = 11/17$.

Normal distribution

One of the most important continuous distribution in statistics.

$$X \sim N(\mu, \sigma^2)$$

Defined in terms of two parameters, the mean μ and variance, σ^2 .

Often parameterised in terms of the mean μ and precision $\tau = 1/\sigma^2$. Thus:

$$X \sim N(\mu, \sigma^2) \equiv N(\mu, 1/\tau).$$

Normal distribution

For $X \sim N(\mu, \sigma^2) \equiv N(\mu, 1/\tau)$, X has probability density function

$$f(x) = \sqrt{\frac{\tau}{2\pi}} \exp\left(-\frac{\tau}{2}(x - \mu)^2\right) \quad (x \in \mathbb{R}).$$

Therefore the likelihood satisfies

$$\begin{aligned} L(\mu, \tau) = f(\mathbf{x}|\mu, \tau) &= \prod_{i=1}^n \sqrt{\frac{\tau}{2\pi}} \exp\left(-\frac{\tau}{2}(x_i - \mu)^2\right) \\ &\propto \tau^{\frac{n}{2}} \exp\left(-\frac{\tau}{2} \sum_{i=1}^n (x_i - \mu)^2\right). \end{aligned}$$

Known variance

For known variance we are interested in $\pi(\mu|\mathbf{x})$.

Mean μ can take any value between $-\infty$ and $+\infty$.

Natural candidate for the prior on μ : $N(\alpha, 1/\beta)$

Then

$$\begin{aligned}\pi(\mu) &= \sqrt{\frac{\beta}{2\pi}} \exp\left(-\frac{\beta}{2}(\mu - \alpha)^2\right) \\ &\propto \exp\left(-\frac{\beta}{2}(\mu - \alpha)^2\right).\end{aligned}$$

Here α and β are hyperparameters:

mean was estimated from observations with total precision β and with sample mean α .

Writing down the likelihood and prior, one can obtain that the posterior distribution has the form

$$\mu|\mathbf{x} \sim N\left(\frac{n\tau\bar{x} + \alpha\beta}{n\tau + \beta}, \frac{1}{n\tau + \beta}\right).$$

$$\alpha = \beta = 0?$$

If we set $\alpha = \beta = 0$, this corresponds to $N(0, 1/0) = N(0, \infty)$ prior.

This is known as an **improper** prior.

In this case:

$$\mu|\mathbf{x} \sim N\left(\frac{n\tau\bar{x} + 0}{n\tau + 0}, \frac{1}{n\tau + 0}\right) = N(\bar{x}, \sigma^2/n).$$

The posterior distribution for μ has mean the sample mean, $\bar{x}$, which is the MLE and variance, σ^2/n , which is the frequentist standard error of the mean.

Thus we return to something familiar!

General Normal (Gaussian) prior

For a general prior, we can write the posterior mean as a weighted mean between the sample mean $\bar{x}$ (MLE) and prior mean.

Let

$$w = \frac{\beta}{\beta + n\tau},$$

then

$$\begin{aligned}\frac{n\tau\bar{x} + \alpha\beta}{n\tau + \beta} &= \frac{n\tau}{n\tau + \beta}\bar{x} + \frac{\beta}{n\tau + \beta}\alpha \\ &= (1 - w)\bar{x} + w\alpha.\end{aligned}$$

Note that as n increases, w gets smaller and approaches 0. In other words, as we get more and more data, the data dominates the prior.

Observe that "posterior precision" = "prior precision" + $n \times$ "precision of each data item".

Example: Bayesian trial

- Consider a **Phase II study**: does a new anti-cancer drug provides any benefits in terms of Response Rate (θ);



- Binary Outcome:** Response vs No Response;

Responded

Not Responded



- How we could design it using a Bayesian approach?

- Decision are made based on the **posterior probabilities for the treatment effect** given the clinical trial data;

For example, the trial is claiming **the efficacy** if

$$\text{Probability} [\theta \geq q_0 \mid \text{Data, Prior}] > c$$

where

- q_0 is (clinically meaningful) effect threshold
- c is probability threshold (chosen to control type I error)
- “Data” is the observed data
- “Prior” is specified prior distribution on the treatment effect.

Bayesian decision-making: Back to the example

- Assume that the planned sample size is $N = 40$ patients;
- The null response rate is $q_0 = 0.30$
- Clinically interesting response rate is $q_1 = 0.50$

The type I error is defined as

$$\text{Probability}\{\text{Probability}[\theta \geq q_0 \mid \text{Data, Prior}] > c \mid \theta = q_0\}$$

For $c = 0.959$,

$$\text{Prob}\{\text{Prob}[\theta \geq 0.3 \mid \text{Data, Prior}] > 0.959 \mid \theta = 0.3\} = 0.03134$$

The power is defined as

$$\text{Probability}\{\text{Probability}[\theta \geq q_0 \mid \text{Data, Prior}] > c \mid \theta = q_1\}$$

Bayesian decision-making: Back to the example

```
# Trial Setting
```

```
N<-40
```

```
p0<-0.30
```

```
p1<-0.50
```

```
# Generate data
```

```
nsims<-100000
```

```
X<-rbinom(n=nsims,size=N,prob=p0)
```

```
# Find posterior probabilities
```

```
post.prob<-pbeta(p0,shape1=X+1,shape2=N-X+1,lower.tail=F)
```

```
# Find Type I error
```

```
c<-0.959
```

```
mean(post.prob>c)
```

```
[1] 0.03289
```

Assurance: power averaged over what we believe

Power is a *conditional* statement — it presumes one particular true effect δ , usually the one we hope for:

$$\text{Pow}(\delta) = \Pr(\text{reject } H_0 \mid \delta).$$

Assurance integrates that conditional over a design prior $p(\delta)$ (O'Hagan et al., 2005):

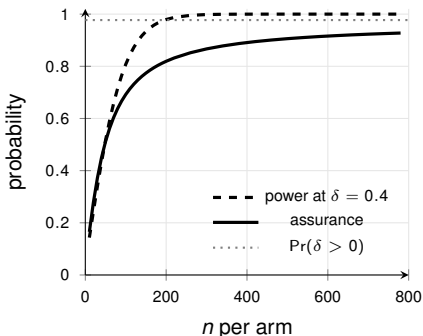
$$\mathcal{A} = \int \text{Pow}(\delta) p(\delta) d\delta = \Pr(\text{trial succeeds}).$$

For a two-arm normal endpoint, n per arm, known σ , one-sided α , and $\delta \sim N(\mu, \tau^2)$, write $k = \sqrt{n/2} / \sigma$:

$$\text{Pow}(\delta) = \Phi(k\delta - z_{1-\alpha}), \quad \mathcal{A} = \Phi\left(\frac{k\mu - z_{1-\alpha}}{\sqrt{1 + k^2\tau^2}}\right).$$

The single number τ does the work: it is the honesty parameter. $\tau \rightarrow 0$ recovers ordinary power at μ .

Assurance has a ceiling



$\delta \sim N(0.4, 0.2^2)$, $\sigma = 1$,
one-sided 2.5%.

- At $n = 100$: power 81%, assurance 69%.
- At $n = 400$: power $> 99.9\%$, assurance 89%.
- As $n \rightarrow \infty$, $\mathcal{A} \rightarrow \Phi(\mu/\tau) = \Pr(\delta > 0) = 97.7\%$.

No sample size buys a success probability above the prior probability that the drug works at all.

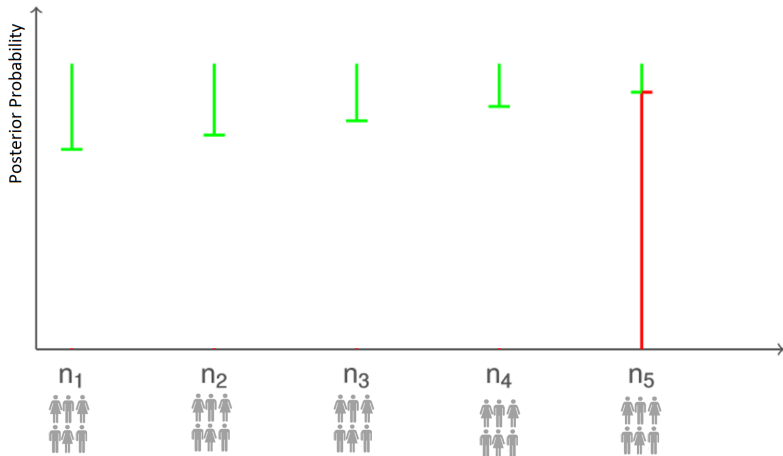
Assurance is not always *lower* than power — at small n the prior's optimistic tail helps. It is lower exactly where it matters: in the region we design in.

Introducing interim analyses

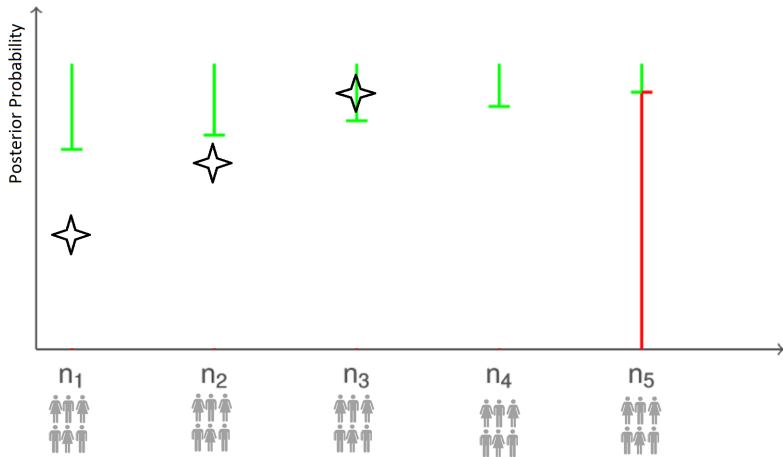
- Assume now we would like to have interim look(s) at the data
- We would like to stop earlier for efficacy if there is early evidence supporting this.
- At interim analysis i with “Data $_i$ ” collected in the trial, we claim efficacy if

$$\text{Probability} [\theta \geq q_0 \mid \text{Data}_i, \text{Prior}] > c_i$$

Group Sequential Trials: efficacy early stopping



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Example: interim analysis

- In our example, assume that we do an interim after 20 patients;
- Early stopping for efficacy only.

What would happen if we keep $c_1 = c_2 = 0.959$?

As we are looking at the data twice, we will increase our chance to make a wrong conclusion (e.g. type I error inflation).

```
X1<-rbinom(n=nsims,size=N/2,prob=p0)
X2<-rbinom(n=nsims,size=N/2,prob=p0)
post.prob.1<-pbeta(p0,shape1=X1+1,shape2=N/2-X1+1,lower.tail=F)
post.prob.2<-pbeta(p0,shape1=X1+X2+1,shape2=N-X1-X2+1,lower.tail=F)
c<-0.959
mean(post.prob.1>c | post.prob.2>c)
[1] 0.0646
```

Need to adjust c to control the type I error

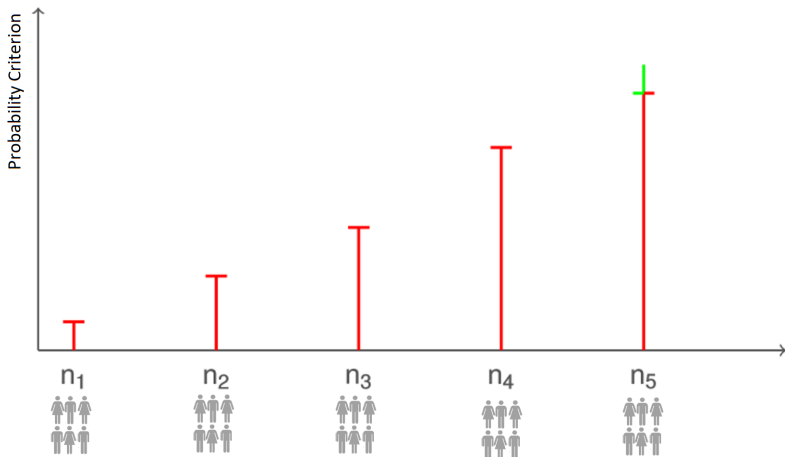
In fact, earlier in the development, it can be more common to stop earlier for futility (lack of effect) rather than efficacy;

- Futility: trial seems unlikely to achieve its objectives
- Futility IA increasingly used in clinical trial design due to ethical motivations, potentially significant savings
- Adding futility leads to power loss;
- “Stopping when we should” versus “Continuing when we should” are always in conflict and need to be balanced

Level of aggressivity of the stopping rule allows to minimize one or the other of the two possible incorrect decisions.

- Aggressive futility rule increases chances of (correct) stopping under null, i.e. reduction in type I error
- but also increase chances of (false) stopping under alternative , i.e. reduction in power
- Cautious futility rule decreases chances of stopping under H_0 but also increase chances of continuing under H_1

Group Sequential Trials



Futility rule based on posterior probability

One way to define the futility stopping is to use the same type of criterion of the posterior probability. Specifically,

- At interim analysis i with “Data $_i$ ” collected in the trial, we stop the trial earlier for futility if

$$\text{Probability} [\theta \leq q_0 \mid \text{Data}_i, \text{Prior}] > t_i$$

- For example, in our example, we would like to stop the trial earlier if the posterior probability that the response rate is below 30% is at least 50%

$$\text{Probability} [\theta \leq 0.3 \mid \text{Data}_i, \text{Prior}] > 0.50$$

Posterior probability futility: example

Assume that after 20 patients, we have observed 5 responses.
Then the probability

$$\text{Probability} [\theta \leq 0.3 \mid \text{Data}_i, \text{Prior}] = 0.637 > 0.50$$

so, we would recommend to stop the trial for futility.

```
> post.prob.futility<-pbeta(p0,shape1=x+1,  
shape2=N/2-x+1,lower.tail=T)  
> post.prob.futility  
[1] 0.6372881
```

Assessing the futility rule

While the probability t_i is informative for the decision-making, it might provide a limited insight into the actual statistical properties of such an early stopping.

What actually matters is the operating characteristics of the futility rule.

- **Power loss (%)**: how much the study power is decreased by imposing a futility rule? (undesirable behaviour);
- **Correct stop under the null (%)**: chance that the futility criterion would be reached under the null (desirable behaviour)

Assessing the futility rule: example

To assess the operating characteristics, we can conduct a simulation study to see how likely we are to stop the trial earlier under different hypothesis (null/alternative). In our example,

```
> post.prob.futility<-pbeta(p0,shape1=X1+1,  
shape2=N/2-X1+1,lower.tail=T)  
> post.prob.efficacy<-pbeta(p0,shape1=X1+X2+1,  
shape2=N-X1-X2+1,lower.tail=F)  
> c<-0.959 # efficacy threshold from single-stage design  
> t<-0.50  # futility threshold  
  
> mean(post.prob.futility<t & post.prob.efficacy>c)  
[1] 0.03118 # Type I error  
  
> mean(post.prob.futility>t)  
[1] 0.41668 # Probability to stop the trial earlier
```

Assessing the futility rule: example

Design Option	Futility Bound	Type I	P(Stop) (under null)	Power	P(Stop) (under alt)
Fixed	—	3.2%	—	78.6%	—
Adaptive	75%	3.2%	23.6%	78.4%	0.6%
Adaptive	50%	3.2%	41.6%	78.6%	2.1%
Adaptive	25%	2.8%	77.4%	74.4%	13.0%
Adaptive (adj)	25%	5.0%	77.5%	80.2%	13.3%

Futility rule based on Conditional Power

Alternative decision-rules are also common to use for the futility stopping. These could be more intuitive to communicate as these directly link to the **predicted success of the trial**.

Conditional Power (CP) is the probability that the trial will reach statistical significance at the final analysis if it continues, given the results at the interim analysis, and assuming **certain response** governs the remainder of the trial.

- The observed treatment effect is assumed for the rest of the trials;
- The treatment effect under the alternative effect is assumed for the rest of the trial;
- If $CP_i \leq I_i$ then we stop the trial earlier for futility
- Choose I_i to achieve desirable OCs

Conditional Power: Example

To achieve our criterion for efficacy at the final analysis, one needs to observe at least 18 responses in 40 patients

```
> x<-18  
> pbeta(p0,shape1=x+1,shape2=40-x+1,lower.tail=F)  
[1] 0.9800707 # > critical value = 0.959
```

Assume that we have observed 9/20 responses at the interim.

We need at least 9/20 more to declare efficacy at the final

Probability to achieve this under the observed rate is 58.6%

```
> sum(dbinom(x=9:20,size=20,prob=9/20))  
[1] 0.5856938
```

Probability to achieve this under 50% (alternative) rate is 74.8%

Conditional Power: Example

To find the critical value for the CP, one can, again, resort to simulations. Specifically, for the considered example

```
CP<-c()
for(i in 1:nsims){
  CP[i]<-sum(dbinom(x=(18-X1[i]):20,size=20,
    prob=(X1[i]/(N/2))))
}
post.prob.efficacy<-pbeta(p0,shape1=X1+X2+1,
shape2=N-X1-X2+1,lower.tail=F)

l<-0.001
> mean(CP>l & post.prob.efficacy>c)
[1] 0.0324 # Type I error
> mean(CP<l)
[1] 0.41393 # Probability to stop early under the null
```

Futility rule based on Predictive Probability of Success (PPoS)

The CP approach allows to assume the response rate for the remainder of the trial. However, the response rate is a **random variable** itself.

- PPoS is the probability that the trial will claim efficacy at the final analysis if it continues, given the results at the interim analysis (and prior) - no further assumptions.
- PPoS is obtained by integrating the data likelihood over the posterior distribution;
- If PPoS is below the futility threshold I_f , then the trial is stopped earlier for the futility

PPoS Futility rule: example

For the binary single-arm trial, we have already come across the predictive distribution! it is the **beta-binomial** distribution

So, instead of computing $\Pr(X = 9 : 20 | \theta = 9/20)$, we compute

$$\Pr(X = 9 : 20 | Data) = \int_0^1 \Pr(X = 9 : 20 | \theta) \times \pi(\theta | Data) d\theta.$$

After 9/20 responses, we need at least 9/20 responses in the second stage to declare efficacy.

The predictive probability to observe these is 57%

```
library("rmutil")  
sum(dbetabinom(9:20, 20, (9+1)/(20+2), 20+2))  
[1] 0.5697892
```

To remind, the CP under the observed effect was 59%/75%.

Comparing different futility rules

At a given time point, a futility rule expressed on any particular scale can be transformed to any other.

These scales are more devices for expressing futility rules rather than meaningful ways of quantifying the chance of trial success.

In choosing a futility criterion, we should focus on operating characteristics.

Assume that there is some **prior knowledge** about the response rate:

1. The most likely response probability is 40%;
2. We are pretty sure (80% sure) that it is between 20% and 65%.

Prior Distribution

Resulting Beta prior distribution is $Beta(3.2, 7.5)$

Design Option	Futility Bound	Type I	P(Stop) (under null)	Power	P(Stop) (under alt)
Uniform	25%	2.8%	77.4%	74.4%	13.0%
$B(3.2, 7.5)$	25%	5.9%	60.9%	84.5%	5.7%
$B(3.2, 7.5)$ (adj)	25%	3.0%	61.0%	77.2%	5.8%

Treatment Effect

- We study the **treatment effect** on the Experimental over the control.
- We measure it via Log Odds Ratio (OR)

$$\theta = \log \left(\frac{p_E \times (1 - p_C)}{(1 - p_E) \times p_C} \right)$$

The trial is stopped **for futility** if

$$\text{Probability} [\text{Log OR} \leq 0.0 \mid \text{Data, Prior}] > 90\%$$

The trial is stopped **for efficacy** if

$$\text{Probability} [\text{Log OR} \geq 0.0 \mid \text{Data, Prior}] > 95\%$$

Discussion

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